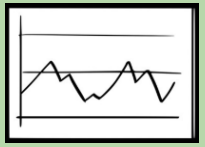
# Research methods



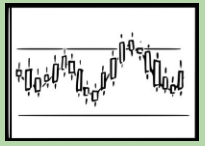
# Plan for today

(1) Topic and supervisor allocations
(2) Cluster training
(3) Literature
(4) Assessment: Sourcing and systematising literature
(5) Questions about your projects answered

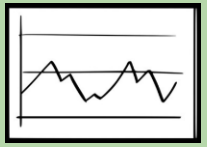
# Topic and supervisor allocations



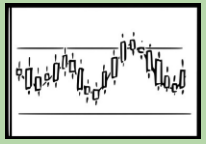
BSc Honours in CAM



BSc Honours in BDA / CS / C



MSc in AI / AIG / CS / C / DS / e / R



MA in e

# Cluster training

MSB has a high-performance computing cluster. If your Honours project will be computationally intensive, you'll probably need to use the cluster.

To use the cluster,

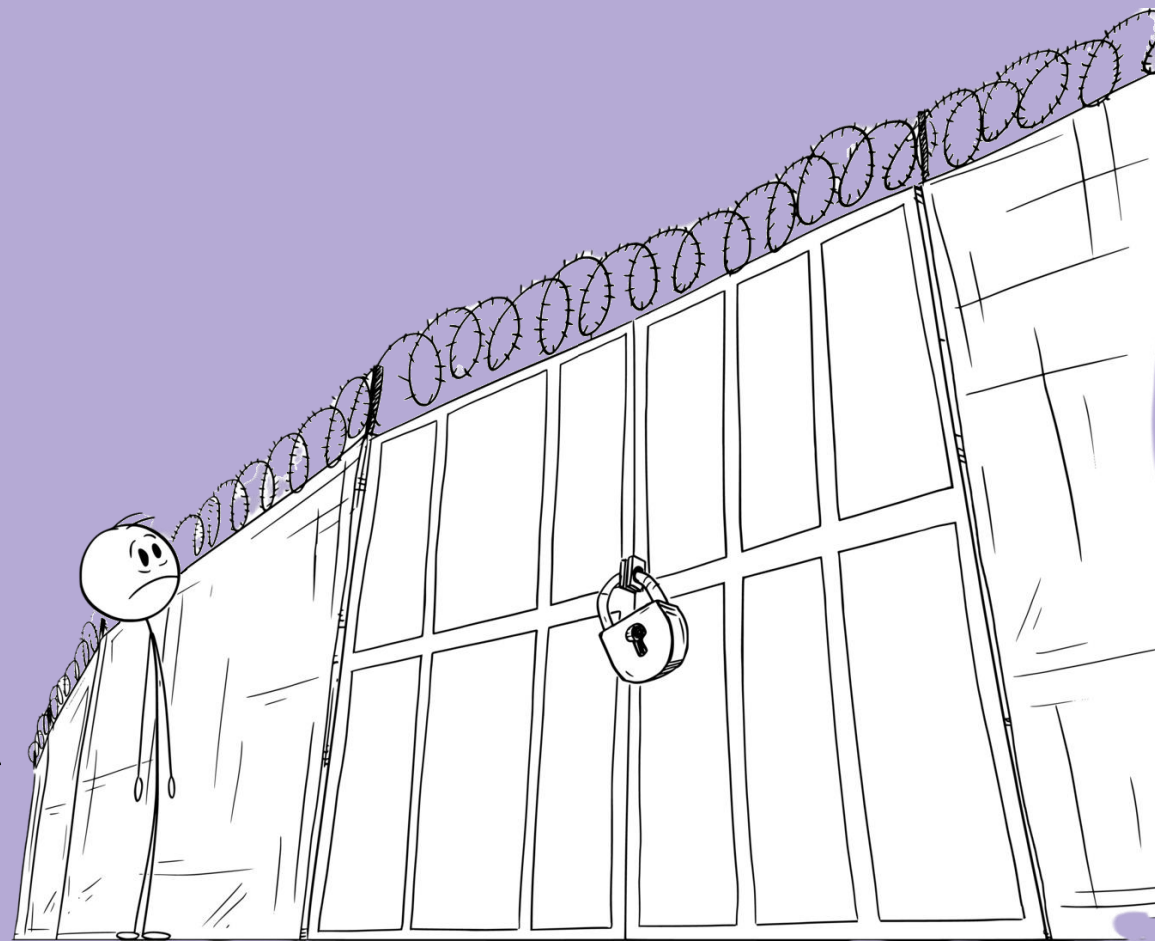
you'll need to complete a set of tasks on Moodle.

To complete the set of tasks on Moodle,

you'll probably need to attend cluster training.

To attend cluster training,

you'll need to complete the Google Form via the link on the IRM/RMCP course site.

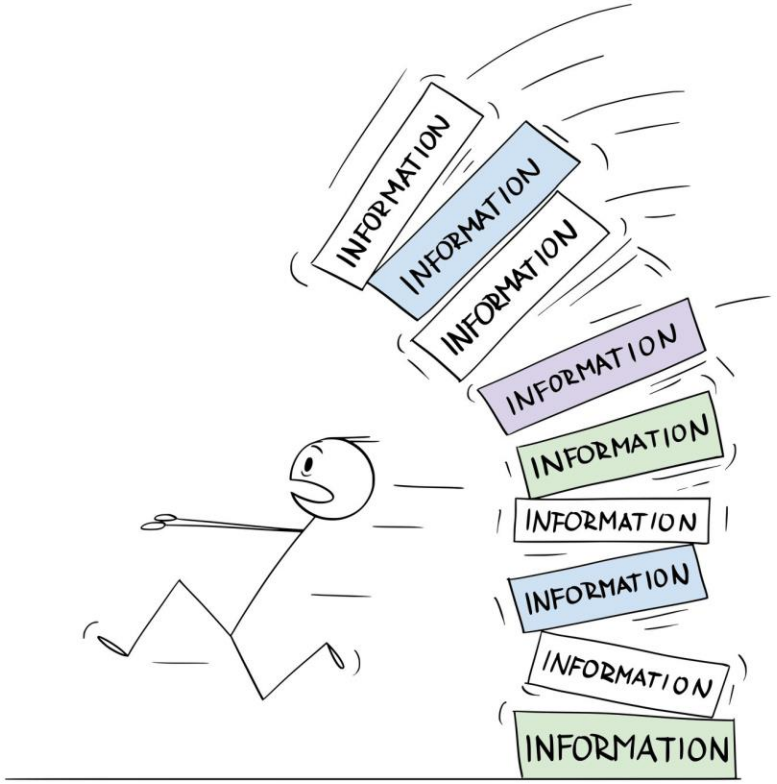


# Literature

For any research project, it's a good idea to begin finding material that's relevant to your area or topic as soon as you can (and then reading or watching or listening to it).

# Literature

There is a lot of material around in a lot of different formats and a lot of different places.



# Literature

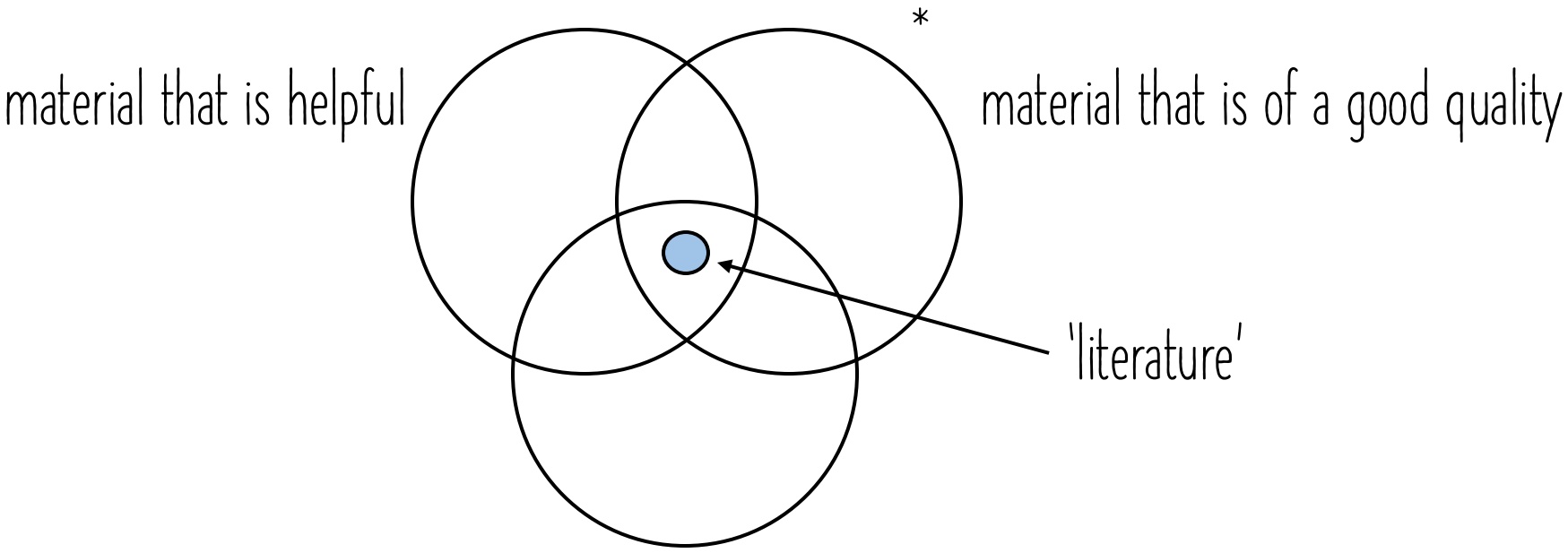
In this lecture, we will discuss what you need from all of that material and how to find it.

# Literature

Here is one important way that all this material can be divided up.

# Literature

Here is one important way that all this material can be divided up.




*not a good Venn diagram

# Literature

In other words, you can use any material that is helpful, of a good quality, and not material that you are supposed to develop yourself, while you are doing your research project. (This is a good idea.)

But, only material that counts as 'literature' will usually appear in the report of your research project.

# Literature



## What is literature?

'Literature' is the subset of all publications that

(1) put forward research and

(2) are recognised by the academic community as being of a sufficiently high quality.

These publications are usually journal articles, conference proceedings, and books.

# Literature



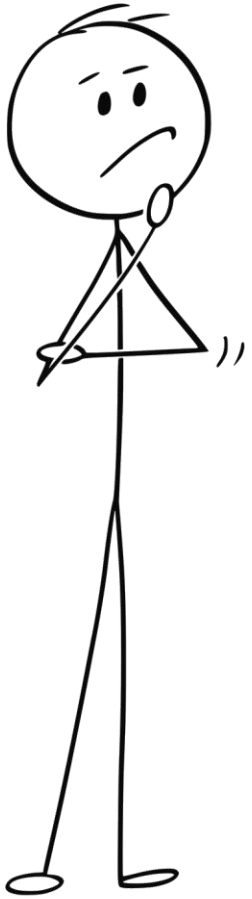
## How do I tell whether some material is literature?

(1) If it is a journal article, you can check whether the journal appears on a list of ranked journals, such as Scimago, Web of Science, or Scopus.

(2) If it is part of a conference proceedings, you can check whether the conference is run by a recognised academic institution or society.

(3) If it is a book, you can check whether it is published by a recognised academic publisher.

# Literature



## How do I find helpful, good-quality literature?

(1) Ask a search engine, like Google. ← not good

(2) Ask an LLM, like ChatGPT. ← possibly better

(3) Ask an academic search engine, like Google Scholar or BASE. ← good

(4) Ask a discipline-specific search engine. ← even better

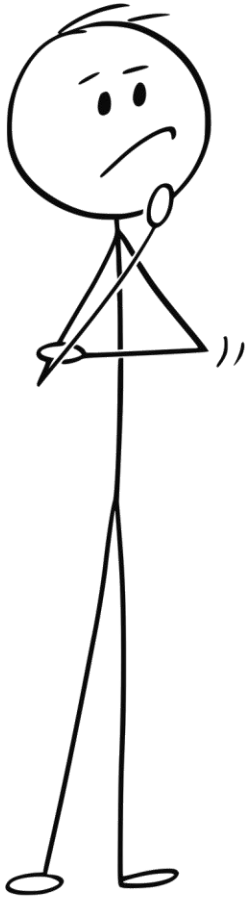
# Literature



## How do I find helpful, good-quality literature?

Remember that your search terms or prompts can make a big difference to what you find.

# Literature



adaptive agents

About 5 470 000 results (**0,12** sec)

**Toward establishing trust in adaptive agents**

A Glass, DL McGuinness, M Wolverton - Proceedings of the 13th ..., 2008 - dl.acm.org

... of **adaptive** assistants; and we discuss recommendations for how to build **adaptive agents** that ... , and the resulting guidelines that these themes imply for designers of **adaptive agents**. ...

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**Adaptive agents**, intelligence, and emergent human organization: Capturing complexity through **agent**-based modeling

BJL Berry, LD Kiel, E Elliott - Proceedings of the National Academy of ..., 2002 - pnas.org

... to the study of complex **adaptive** systems have offered researchers in ... **adaptive** systems are characterized often by "**agents**" ... fit the description of a complex **adaptive** system. The difficulty ...

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learning agents

About 7 460 000 results (**0,14** sec)

**Simulation with learning agents**

E Gelenbe, E Seref, Z Xu - Proceedings of the IEEE, 2001 - ieeexplore.ieee.org

... We propose that **learning agents** (LAs) be incorporated into simulation environments in ... may receive from other **agents**. We investigate an approach in which **agents** are assigned goals ...

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**Learning agents** for uncertain environments

S Russell - ... eleventh annual conference on Computational **learning** ..., 1998 - dl.acm.org

... " for a **learning agent** that can handle stochastic, partially observable environments. The architecture uses reinforcement **learning** ... we can try out the complete **agent**; more arise when we ...

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**Learning** to incentivize other **learning agents**

J Yang, A Li, M Farajtabar, P Sunehag... - Advances in ..., 2020 - proceedings.neurips.cc

... Instead, we allow **agents** to **learn** an incentive function that gives ... **learning agents** and thereby shape their behavior. The new **learning** problem for an **agent** becomes two-fold: **learn** a ...

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# Literature



## How do I find helpful, good-quality literature?

If you already have some literature, you can use this to help you with search terms and also to find other related literature.

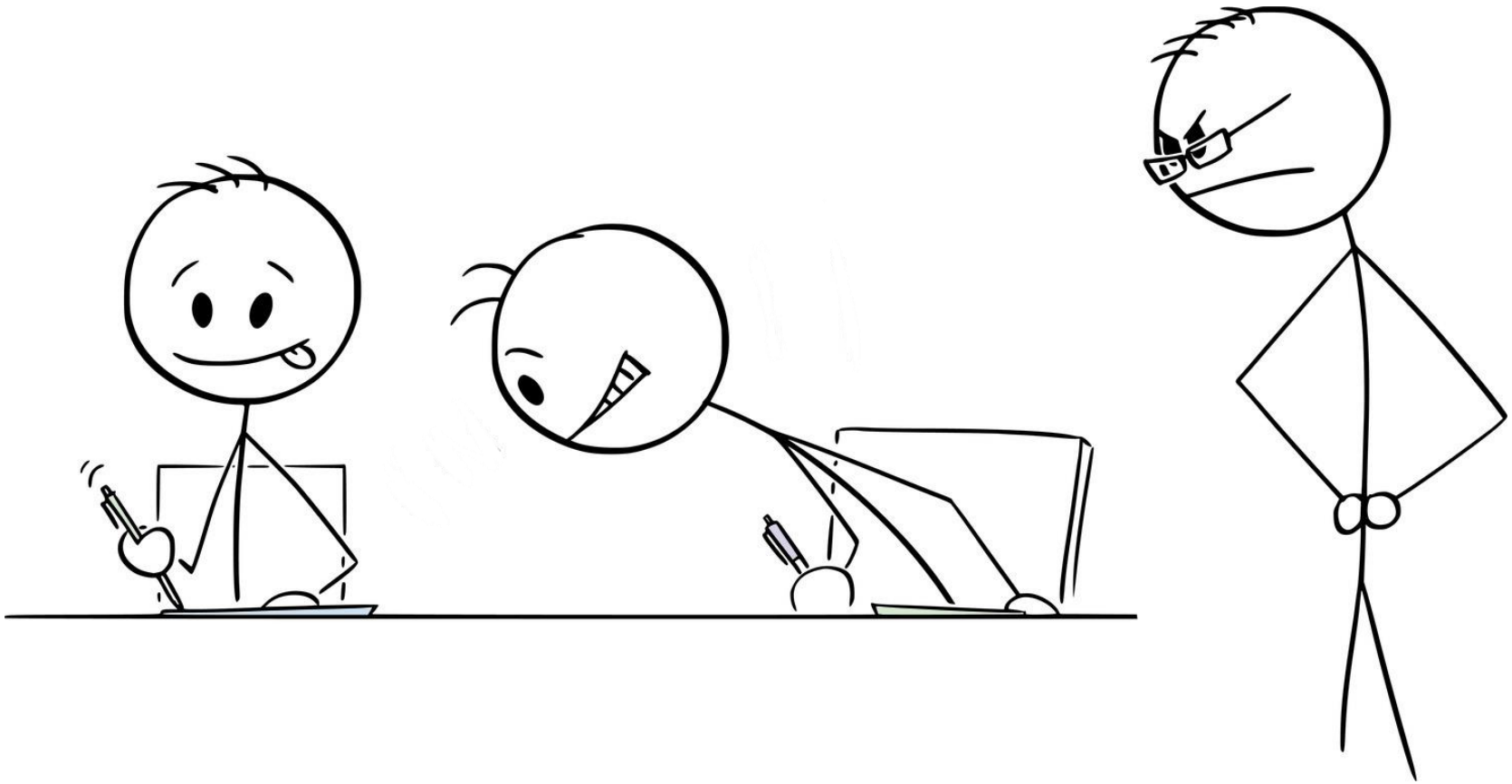
# Literature



## How much literature should I find?

This is relative to
(1) the level of your project and
(2) how broad or narrow your area or topic is.

# Assessment: Sourcing and systematising literature

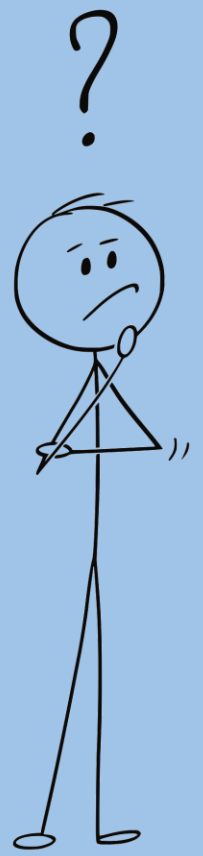


# Assessment: Sourcing and systematising literature

The purpose of this assessment is for you to begin sourcing, reading, and systematising literature that is relevant to your project.

The format of this assessment will be chosen by your supervisor(s). Your supervisor(s) could choose for you to present the literature as an annotated bibliography, for you to present the literature using slides during one of your meetings, or for you to present it in some other way.

# Questions about your projects answered





# Thank you for attending the lecture.